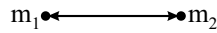


Gravitation

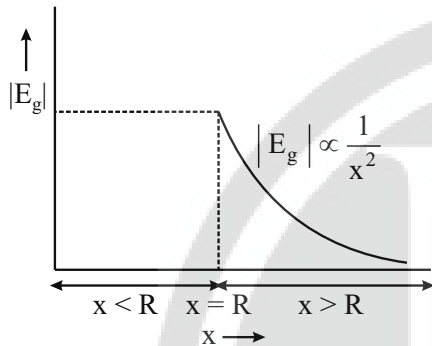
SHORT NOTES

Newton's Universal Law of Gravitation



- Force of attraction between two point masses $F = \frac{Gm_1m_2}{r^2}$, where $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
- Directed along the line joining the point masses.
- It is a conservative force $\Rightarrow$ mechanical energy will be conserved.
- It is a central force $\Rightarrow$ angular momentum will be conserved.

Gravitational Field due to Spherical Shell



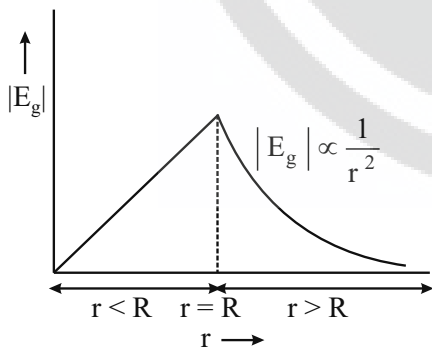
Outside Region $E_g = \frac{GM}{r^2}$, where $x > R$

On the surface $E_g = \frac{GM}{R^2}$, where $x = R$

Inside Region $E_g = 0$, where $x < R$

Note: Direction always towards the centre of the sphere, radially inwards.

Gravitational Field Due to Solid Sphere



Outside Region $E_g = \frac{GM}{r^2}$, where $r > R$

On the surface $E_g = \frac{GM}{R^2}$, where $r = R$

Inside Region $E_g = \frac{GMr}{R^3}$, where $r < R$

Gravity 'g'

- Acceleration due to gravity $g_s = \frac{GM}{R^2}$ (on the surface of earth)
- At height h , $g_h = \frac{GM}{(R+h)^2}$
- If $h \ll R$; $g_h \approx g_s \left(1 - \frac{2h}{R}\right)$
- At depth d , $g_d = \frac{GM(R-d)}{R^3} = g_s \left(1 - \frac{d}{R}\right)$
- Effect of rotation on g : $g' = g - \omega^2 R \cos^2 \lambda$ (where λ is angle of latitude.)

Gravitational Potential

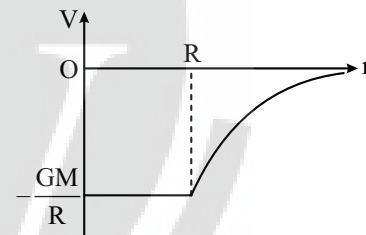
- Due to a point mass at a distance r

$$V = \frac{GM}{r}$$

- Gravitational potential due to spherical shell

Outside the shell $V = \frac{GM}{r}$, $r > R$

Inside/on the surface of the shell $V = \frac{GM}{R}$, $r < R$

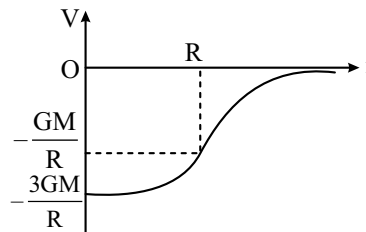


- Potential due to a solid sphere

Outside Region $V = -\frac{GM}{r}$, $r > R$

On the surface $V = -\frac{GM}{R}$, $r = R$

Inside Region $V = -\frac{GM(3R^2 - r^2)}{2R^3}$, $r < R$



- Potential on the axis of a thin ring at a distance r from the centre

$$V = -\frac{GM}{\sqrt{R^2 + r^2}}$$

Motion of a Satellite

- ❖ Escape velocity from a planet of mass M and radius R

$$v_e = \sqrt{\frac{2GM}{R}}$$

- ❖ Orbital velocity of satellite (orbital radius r)

$$v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{(R+H)}}$$

- ❖ For nearby satellite

$$v_0 = \sqrt{\frac{GM}{R}} = \frac{v_e}{\sqrt{2}}$$

Here V_e = escape velocity on earth surface.

Time Period of Satellite

$$T = \frac{2\pi r}{v} = \frac{2\pi r^{3/2}}{\sqrt{GM}}$$

Energies of a Satellite

$$\text{Potential energy } U = -\frac{GMm}{r}$$

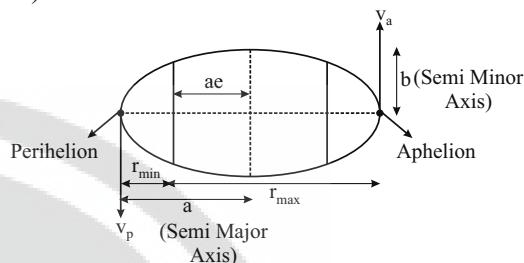
$$\text{Kinetic energy } K = \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

$$\text{Mechanical energy } E = U + K = -\frac{GMm}{2r}$$

$$\text{Binding energy } BE = -E = \frac{GMm}{2r}$$

Kepler's Laws

- ❖ 1st Law of orbitals Path of a planet is elliptical with the sun at one of the focus.
- ❖ IInd Law of areas Areal velocity $\frac{d\vec{A}}{dt} = \text{constant} = \frac{\vec{L}}{2m}$
- ❖ IIIrd Law of periods: $T^2 \propto a^3$ or $T^2 \propto \left(\frac{r_{\max} + r_{\min}}{2}\right)^3 \propto (\text{mean radius})^3$ For circular orbits $T^2 \propto R^3$



- ❖ $r_{\max} = a(1 + e)$
 $r_{\min} = a(1 - e)$
- ❖ $\frac{v_p}{v_a} = \frac{v_{\max}}{v_{\min}} = \frac{1+e}{1-e}$ (where, e is eccentricity.)